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In this homework you will apply finite difference methods on rectangular and non-rectangular 2D domains using MATLAB (or any other programming language of your choosing). You will perform numerical the solution of Poisson's equation using finite difference methods for a 2D electric potential. **Don't worry too much if electric potential/electric fields are not something you are familiar with – we will focus mostly on the math/numerical methods.**

1 Background: Problem Inspiration

This HW problem is inspired by the biological process of **electrolocation** [1] – a process used by *weakly electric* fish to detect nearby objects in water (and also to communicate). These fish (e.g., the elephantnose fish, *Gnathonemus petersii*) use an electric organ discharge (EOD) located in their tail to generate a pulse that creates an electric potential and electric field in the surrounding water. Electroreceptors (e.g., Ampullae of Lorenzini) distributed on the fish's body can then detect the electric field and “sense” any changes from one pulse to the next caused by the presence or appearance of nearby objects. This is similar to the echolocation process used by bats, but in this case, electric fields are used for the sensing process.

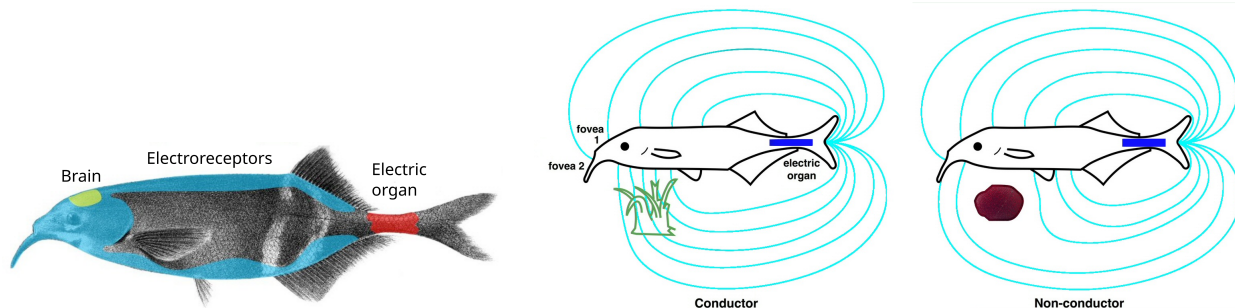


Figure 1: Images from Wikipedia. (Left) [9] The elephantnose fish (*Gnathonemus petersii*) is a weakly electric fish that uses electrolocation to detect nearby objects in the murky water and (right) [10] A depiction of the electric field generated by the charge source in the fish's tail interacting with nearby objects.

There have been several research efforts to understand and model weakly electric fish's electrolocation – if you review these research papers/dissertations you will find that the modeling required

to understand fish electrolocation is quite a bit more complicated than the (very) simplistic model we use in this homework [2, 3, 4, 5, 6, 7, 8]. This said, hopefully you find this assignment interesting and fun.

2 Description of Simplified PDE Model and the Physical Geometry

For this homework, we greatly reduce the complexity by simplifying both the geometry as well as the mathematical model of the physical system. **The result is that we will model a problem inspired by weakly electric fish to understand conceptually how *electric imaging* works. This is not the best mathematical model to describe the complexity of weakly electric fish electrolocation.**

The following problem description and assumptions are employed in this homework:

- **Problem goal:** our goal will be to examine and assess how well an electric field can sense a nearby “wall” – whether the distance to the wall and or the slope of the wall is detectable. Understanding this “electro-sensing” capability might be useful for developing a robotic sensor inspired by this biological process.
- **Physical Domain & Governing Equation(s):** We will use a simplified physical domain for this problem – see the figure below. Some relevant information about the domain:
 - The electric potential in the domain will be determined by numerically solving the Poisson equation – which models the static electric potential behavior in the domain:

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0, \quad (1)$$

Once the electric potential is determined, the electric field (a vector field) can be calculated by post-processing the gradient of the potential:

$$\nabla \phi = -\vec{E}, \quad (2)$$

- The bottom boundary of the physical domain ($\Gamma_{d-e-O-f-a}$) is a “mirror plane” – meaning that the true physical domain is reflected across the x-axis. We will only solve the “top half” of the problem as shown in the domain. To this end, on portions of the domain that are not generating electric field, the $\frac{\partial \phi}{\partial n} = 0$
- The electric field generator exists on the bottom boundary and will be modeled as a 4-unit long line with prescribed electric potential (Voltage), with $\phi = -1$ on Γ_{E-O} , $\phi(0,0) = 0$ at Γ_O and $\phi = 1$ on Γ_{O-f} as shown in the domain figure above.
- Because we would want to test our method in a controlled experiment, the top and side boundaries of the domain will be modeled as “impermeable walls” which will be represented using the boundary condition: $\frac{\partial \phi}{\partial n} = 0$, applied to Γ_{a-b} , Γ_{b-c} , and Γ_{c-d} .

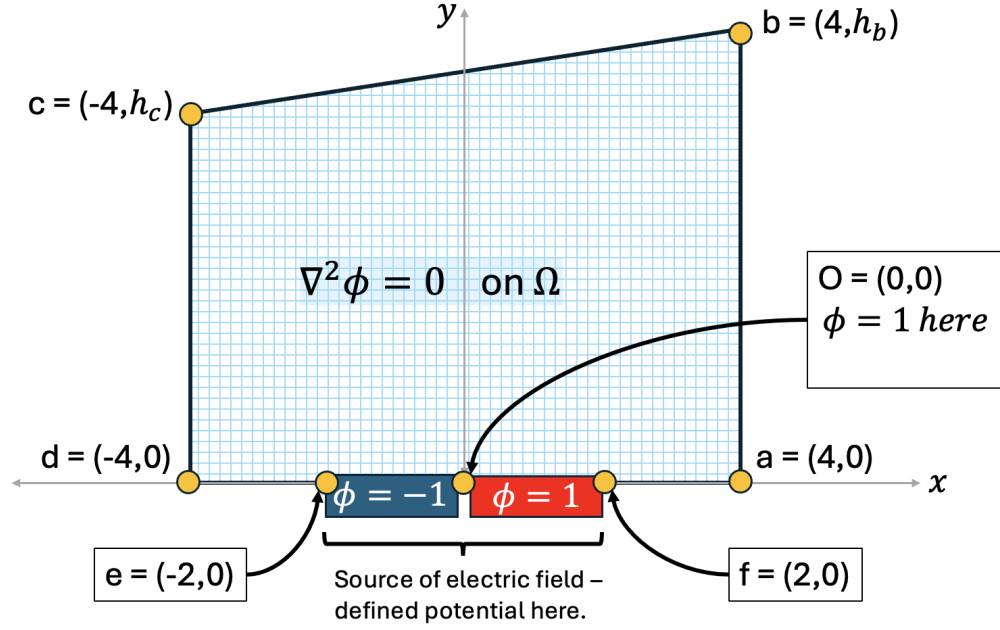


Figure 2: The physical domain with the model electric field generator shown at the bottom.

- The sides of the domain (Γ_{a-b} , and Γ_{c-d}) are perpendicular to the bottom of the domain, but will vary in height during our numerical study of the sensor in question 9 (heights defined by h_b and h_c).

3 Homework # 1 Questions and Deliverables

Please answer the following questions. Please submit all answers electronically on GRADESCOPE as a single PDF. Written answers and diagrams performed on paper may be scanned and merged with typed answers/code. You do not need to type all of your work. Plots and calculated results should be included in the report. Any computer code used in completing the assignment should be included in the submission.

Please note: If you are unable to solve the problem/implement the code as prescribed, you may submit a simpler version of the problem for partial credit (e.g., if you are having difficulty with the mapping, you may submit a rectangular domain solution, or if you are having difficulty with the boundary conditions, you may solve the problem with Dirichlet boundary conditions on all boundaries, etc.).

If you choose to use MATLAB, exporting a Live Script directly to a PDF is the preferred method for submitting your deliverable..

3.1 Academic Integrity-Please include this statement on your submission coverpage and sign

You MAY use MATLAB, your class notes, YouTube videos, and the "static" internet (static here means: not communicating/asking questions that would be answered by a person via the internet – inclusive of posting to forums).

You MAY NOT receive help from any people external to this class (ie. you may not post to forums online, internet answer sites such as CHEGG).

You must submit your own work and your own code. If you do discuss and or receive help from another person in the class, you must state what help/assistance you received as well as have that person sign to agree that was the extent of the assistance/help. If you use external resources such as generative AI as well as any references, please be sure to clearly cite their use and list how it was used/contributed to your homework.

1. (5 pts) **Bilinear Mapping:** On paper describe the bilinear mapping functions, $x(\xi, \eta)$ and $y(\xi, \eta)$. Clearly show the matrix system used to find the unknown mapping coefficients with h_c and h_b to define the mapping. Provide evidence (MATLAB plots) that the mapping works.
2. (5 pts) **Finite difference approximations of derivatives:** List ALL of the finite difference equations **with truncation error** $O(\Delta x^2)$ that will be used to approximate the partial derivatives in the numerical approximation of the governing PDE that will be solved in the computational domain.
3. (10 pts) **Internal Domain Nodes: Finite difference representation of the governing equation** Write the finite difference equation that will be used to represent the governing equation that will be satisfied at each internal node in the computational domain. Please also draw the associated 9-point finite difference stencil. Create a 3×3 table containing the coefficients (AA, BB, CC, etc) for each point used in the finite difference stencil.
4. (20 pts) **Pseudo-code description:** Write a detailed pseudo-code that describes the MATLAB process of setting up and solving the finite difference problem. Please note, you should also include any post-processing that you performed to calculate the electric field.
5. (10 pts) **Boundary Conditions:** Describe clearly on paper how you would introduce the boundary conditions for a non-rectangular domain which will be used.
6. (10 pts) **Postprocessing – Electric Field Calculation:** Describe clearly on paper how you will use finite difference equations to perform the post-processing calculation for the electric field. Hint: you will need to calculate $E_x = \frac{\partial \phi}{\partial x}$ and $E_y = \frac{\partial \phi}{\partial y}$ for each node.
7. (20 pts) **Programming a working solver:** Following the example from class or starting from scratch, write your own finite difference solver to solve the Laplace/Poisson equation for

this problem with the *boundary conditions provided in the description implemented*. Please include an implementation of your proposed method to calculate the electric field. *If you did not get your solver working, please describe what was the main issue you faced. Also, provide a clear description of any simplifications you made to this HW so that you could solve the problem.*

8. (10 pts) **Demonstrate Solution of Potential field, Electric field, and solver Convergence:** Demonstrate that your finite difference solver works by providing a contour plot of the potential field and a 2D quiver plot of the electric field on a non-square mesh. Briefly describe (in written text) how you demonstrated convergence with mesh refinement. Show the order of convergence of your PDE finite difference solver as a function of grid size. If your convergence rate differs from $O(\Delta x^2)$, explain briefly why.
9. (10 pts) **Numerical Study of a simple electrolocation sensor:** In this final problem, you will use our numerical tool to examine whether electrolocation can be used to identify (a) the distance to a wall as well as (b) the slope of the wall. You should perform a systematic numerical study (use a medium refined mesh) in which you vary the distance from the electric field source to the top domain boundary and vary the angle of the boundary (ie. you should vary h_c and h_b). You should consider using an approach in which you use/examine the electric field at the lower boundary of the domain (where the electric field generator is) in the normal direction to electric field generator or EOD. By observing changes in the electric field, you should be able to come up with a “relationship” or contour plot that shows how your sensed quantity changes as you change the slope and distance to the wall.

References

- [1] Angel Ariel Caputi; Living life with an electric touch. J Exp Biol 1 December 2023; 226 (23): jeb246060. doi: <https://doi.org/10.1242/jeb.246060>
- [2] H. Ammari and T. Boulrier and J. Garnier and H. Wang, Mathematical modelling of the electric sense of fish: the role of multi-frequency measurements and movement, Research Report No. 2017-11 February 2017, Seminar für Angewandte Mathematik Eidgenössische Technische Hochschule CH-8092 Zürich Switzerland, accessed 2/6/2025: https://www.sam.math.ethz.ch/sam_reports/reports_final/reports2017/2017-11_fp.pdf
- [3] Assad, C. (1997). Electric field maps and boundary element simulations of electrolocation in weakly electric fish. PhD thesis, California Institute of Technology, Pasadena, CA. University Microfilms Inc., Ann Arbor, MI.
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- [5] Babineau, D. (2006). Modeling the electric field and natural environment of weakly electric fish. MSc thesis, University of Ottawa, Ottawa, ON, Canada.
- [6] Bacher, M. (1983). A new method for the simulation of electric fields, generated by electric fish, and their distortions by objects. *Biol. Cybern.* 47, 51-58.
- [7] Rasnow, B. (1996). The effects of simple objects on the electric field of *Apteronotus*. *J. Comp. Physiol. A* 178,397-411.
- [8] Rasnow, B., Assad, C., Nelson, M. E. and Bower, J. M.(1989). Simulation and measurement of the electric fields generated by weakly electric fish. In *Advances in Neural Information Processing Systems* (ed. D. S. Touretzky), pp.436-443. San Mateo, CA: Kaufmann Publishers.
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- [10] https://en.wikipedia.org/wiki/Electroreception_and_electrogenesis#/media/File:Scene_analysis_in_electroreception_coloured.jpg